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Do Lai Tin Dat

AlGaN/GaN-Based Transistor Development

Ohmic Contact and Schottky Gate Contact Characterization

SUPERVISOR

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1. Introduction

The recent demand for faster, more powerful, and more reliable electronic systems has driven the search for semiconductor materials capable of operating beyond the limits of conventional Silicon technology. AlGaN/GaN high-electron-mobility transistors (HEMTs) have gained significant attention due to the exceptional properties of the III-V nitride material system, and most notably, the spontaneous formation of a two-dimensional electron gas (2DEG) at the AlGaN/GaN heterointerface. These properties have made AlGaN/GaN HEMTs available for high-power and high-frequency applications, including 5G base stations, radar systems, satellite communications, and aerospace electronics, as well as for operation in extreme environments characterized by high temperatures and intense radiation [1]. The properties will be discussed in detail in Chapters 2.1 and 2.2.

The performance of AlGaN/GaN HEMT devices is strongly dependent on the quality of both the ohmic contacts at the source and drain, and the gate contact, which controls the 2DEG channel. Due to the inherent formation of conductive 2DEG channel, these devices conduct without any applied gate voltage – they are called normally-on devices. While normally-off HEMTs offer advantages such as lower power consumption and simplified circuit design due to their positive threshold voltage, many recent normally-off approaches still face challenges including threshold voltage instability and reduced current density [1], [2]. Depletion-mode (normally-on) HEMTs with Schottky gate contacts therefore remain preferred for high frequency, low voltage applications due to their simpler and more mature fabrication process [ref]. The quality of the Schottky gate contact thus directly determines the RF performance of the device, motivating a detailed investigation of its electrical parameters. Schottky contact and ohmic contacts will be discussed in Chapters 2.4 and 2.5.

In this project, the Schottky gate contact of RF AlGaN/GaN HEMTs is characterized using circular transmission line model (CTLM) test structures, and two parameter extraction methods – including the thermionic emission (TE) extraction method, and its modified version, the Cheung's method – are applied and compared. The ohmic contact is also characterized using the standard TLM method. The measurement setup, results, and discussion will be presented in Chapter 3.

2. Fundamentals of AlGaN/GaN HEMTs

2.1. Material properties

GaN is a wide-bandgap semiconductor with an energy gap of 3.4eV, significantly larger than those of conventional semiconductors such as Si (1.1eV) and GaAs (1.42eV), and comparable to SiC (3.3eV) [3]. This wide bandgap allows the GaN-based devices to operate reliably at extremely high temperatures (above 400-500°C) and withstand the radiation-induced ionization effect [2].

GaN has a high breakdown field (3.3 MV/cm) – 11 times higher than that of Si (0.3 MV/cm) – enabling devices to operate under high voltages, which is essential for high power applications [2]. Furthermore, for the same applied voltage, a GaN layer can be made 11 times thinner than Si layer, reducing significantly resistive losses in switching mode power supplies.

A unique property of AlGaN/GaN heterostructures is that a high-mobility channel of up to 2000 cm^2/Vs [2] can be obtained through the inherent formation of a 2DEG at the interface without any intentional doping, unlike conventionally doped semiconductors, where donor atoms are present directly within the conducting channel and reduce carrier mobility through ionized impurity scattering. Combined with a large electron saturation velocity of 2.5×10^7 cm/s [2], these properties enable AlGaN/GaN HEMTs to deliver outstanding high-frequency performance. The formation mechanism of the 2DEG will be discussed in detail in the following chapter.

2.2. Formation of 2DEG

A distinctive feature of the AlGaN/GaN material system is the presence of strong internal polarization. There are two polarization mechanisms. First, spontaneous polarization (P_{sp}), originating from the asymmetric wurtzite crystal structure and the electronegativity difference between Ga and N atoms, which exists in both GaN and AlGaN independently. The magnitude of P_{sp} in the AlGaN layer depends on the Al composition and differs from that of the underlying GaN layer. Second, piezoelectric polarization (P_{pz}), which arises from the tensile strain in the AlGaN layer due to the lattice mismatch between the two layers.

Since the total polarization in AlGaN and GaN differ, a net positive polarization charge appears at the AlGaN/GaN interface, creating an electric field across the AlGaN barrier. In terms of the energy band diagram, the conduction bands of both AlGaN and GaN bend downward toward the interface, forming a triangular quantum well.

However, the origin of the 2DEG electrons in undoped structures has been a subject of debate. The most widely accepted explanation is the surface donor state model [4], in which donor states on the AlGaN surface ionize and release electrons, which are then swept to the interface by the polarization-induced electric field and confined in the quantum well – forming the high-density, high-mobility 2DEG.

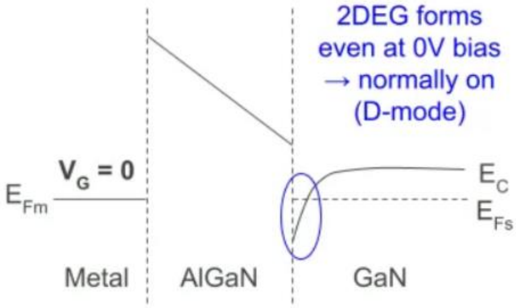

Figure 1. Energy band gap of AlGaN/GaN heterostructure at zero bias

2.3. Device Structure and Working Principle

The AlGaN/GaN HEMTs in this work is fabricated on a purchased epitaxial wafer (NTT-AT, NM140819-1), consisting of a p-type Si (111) substrate, a buffer layer, a 1000 nm intrinsic GaN channel layer, a 25 nm intrinsic $Al_{0.25}Ga_{0.75}N$ barrier layer, and a 3 nm intrinsic GaN cap layer. This cap layer reduces surface defects that would act as electron traps – capturing electrons from the surface donor states before they reach the 2DEG channel, an effect sometimes referred to as the "virtual gate" effect. Fortunately, due to its very thin, the effect of the cap layer on the Schottky gate contact is negligible and will not be considered in the gate characterization presented in Chapter 3. The choice of substrate, buffer layer structure, and Al composition in the barrier layer significantly affects device characteristics; however, a detailed investigation of these parameters is beyond the scope of this work. Then, the wafer was processed in-house involving mesa isolation etching for active regions, ohmic deposition at the source and drain contacts, and the Schottky gate contact deposition.

The HEMTs are majority carrier devices (electrons in this work) and operate as three-terminal field-effect transistors (FETs), where the current flows through the 2DEG channel between source and drain, and is controlled by the gate voltage. The channel conductivity is given by $\sigma = q \cdot n_s \cdot \mu$, where q is the electron charge, n_s is the 2DEG sheet carrier concentration, and μ is the electron mobility. At zero gate bias, the 2DEG is fully formed and the device conducts, it is a normally-on device. Applying a negative gate voltage creates a depletion region beneath the gate, reducing the 2DEG density and therefore the drain current, until the channel is fully pinched off and the device turns off. Above the pinch-off voltage, the 2DEG channel is fully open and the drain current saturates at its maximum value I_{DSS} with increasing drain-source voltage.

Since the gate directly controls the 2DEG channel, its electrical quality is critical to overall device performance. This motivates us to have parameters to describe the control quality of the gate of the examined device. While transconductance $g_m = \partial I_{DS}/\partial V_{GS}$, where I_{DS} drain current and V_{GS} gate voltage, is a well-known parameter describing the gate control efficiency at the device level, it does not reveal much about the fundamental physical properties of the metal/semiconductor junction, which are essential for understanding and optimizing the fabrication process. These contact-level parameters will be discussed in detail in next Chapter.

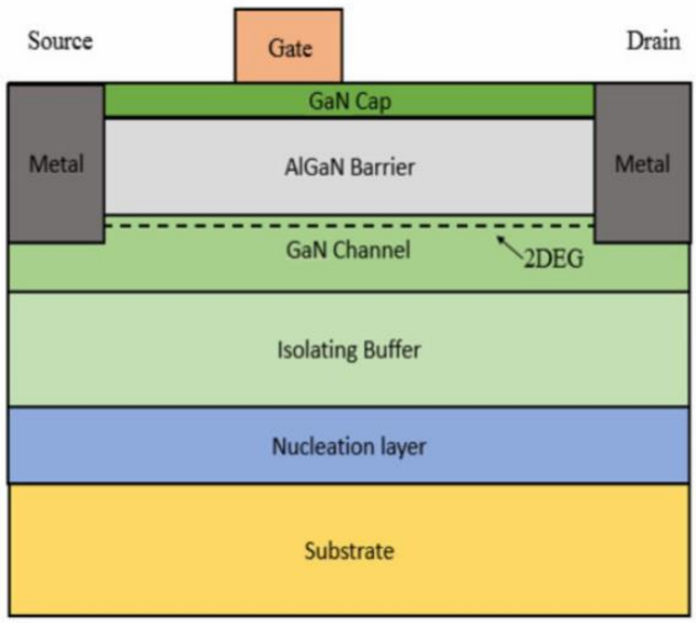

Figure 2. Cross section of an AlGaN/GaN HEMT

3. Characterization of the Contacts

3.1. Ohmic Contact

Ohmic contact is a metal/semiconductor junction that allows the current to flow freely in both directions. In the context of AlGaN/GaN HEMTs, ohmic contacts at the source and drain terminals are essential for ensuring efficient carrier injection between the metal electrodes and the 2DEG channel. The ohmic contact is characterized by the contact resistance R_C (Ω) – the total resistance of the metal/semiconductor junction, which depends on contact geometry, and specific contact resistance ρ_c ($\Omega \cdot mm^2$) – which is independent of the contact geometry. In the lateral devices, the contact resistance is also commonly expressed in unit of $\Omega \cdot mm$, which is independent of device width but depends on the transfer length. These values should be minimized to reduce the parasitic voltage drop across the source and drain contacts, and the better the overall device performance in terms of on-resistance, cut-off frequency and maximum frequency [5].

When a metal is brought into contact with a semiconductor, a Schottky barrier forms at the interface due to the difference in work functions between the two materials. According to Schottky-Mott theory, the metal/semiconductor Schottky barrier height (SBH) for an n-type semiconductor (due to the AlGaN barrier layer is unintentionally n-doped [2]) is given by

$$\Phi_B = \Phi_m - \mathrm{X}_s/q \qquad (1)$$

where Φ_m is the work function of the metal, and X_s is the electron affinity of the semiconductor. For current to flow freely across the junction, electrons must either overcome this barrier through one or more transport mechanisms: thermionic emission, in which electrons gain sufficient thermal energy to get over the barrier; tunnelling (field emission), in which electrons quantum-mechanically pass through a sufficiently thin barrier; and thermionic field emission, in which above both mechanisms occur simultaneously. These transport mechanisms suggest two fundamental approaches to forming ohmic contact structure: reducing the SBH by selecting a metal with a low work function; and thinning the depletion region by increasing the local n-type doping concentration beneath the contact, which narrows the barrier sufficiently for electrons to tunnel through directly.

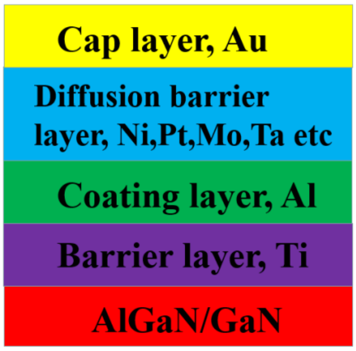

Figure 3. Ti/Al/X/Au Ohmic metallization schemes

Various ohmic contact metallization schemes have been reported for AlGaN/GaN HEMTs; however, the Ti/Al/X/Au scheme is widely recognized as reliable [6]. In this work, a Ti/Al/Ti/Au metal stack is used. The choice of this alloy scheme can be understood qualitatively through the two approaches described above [6]. The Ti and Al layers are well-known for their low work functions (Ti: 4.33 eV, Al: 4.28 eV), and during rapid thermal annealing, the formation of TiN further lowers the effective work function and generates nitrogen vacancies that locally increase the n-type doping concentration beneath the contact. The first Ti layer also acts as a barrier preventing Al from diffusing into the epitaxial layers, while the Al layer additionally prevents oxidation of Ti and contributes low resistivity. The Au cap layer provides a low resistivity external connection (e.g. wire boding) and reduces the overall contact resistivity. A second Ti diffusion barrier layer is inserted between Al and Au to prevent the fast diffusion of Au into the Al layer. It should be noted that the thickness of each layer, their ratios, and the annealing parameters such as temperature and duration significantly affect the ohmic contact characteristics; however, a detailed investigation of these effects is beyond the scope of this work.

3.1.1. Transmission Line Model (TLM)

Since the contact resistance R_C cannot be directly measured from a single contact, a dedicated test structure – the transmission line model (TLM) – is used to extract it independently from the sheet resistance of the semiconductor beneath [7]. The TLM structure consists of a series of rectangular metal pads of width Z and length L, placed with increasing gap spacings d on a mesa-isolated semiconductor strip of width W, as shown in below figure.

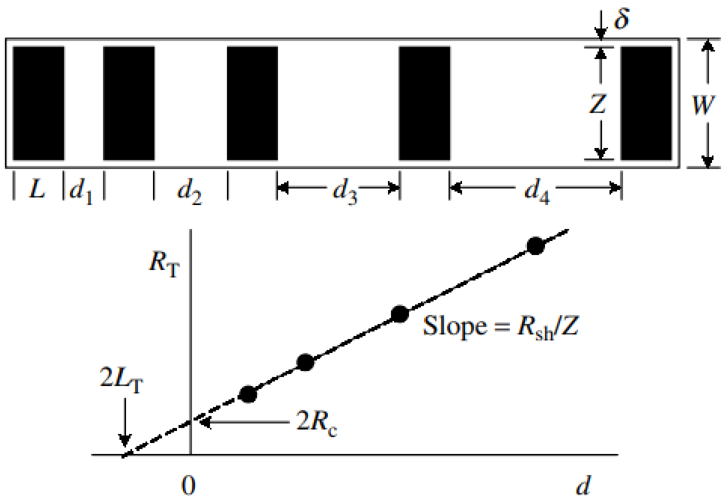

Figure 4. TLM structure and total resistance - spacing plot [7]

The total resistance between any two adjacent pads is measured either by 2-terminal or 4-terminal measurement. The total resistance between any two adjacent pads is given by

$$R_{total} = 2 \cdot R_C + R_{sh} \cdot \frac{d}{Z} \qquad (2)$$

where R_{sh} is the sheet resistance of the semiconductor beneath the gap. Since R_{total} varies linearly with gap spacing d, plotting R_{total} against d yields a straight line, from which two key parameters are extracted directly: the sheet resistance R_{sh} from the slope, and the contact resistance R_C from the y-intercept. However, to obtain the geometry independent specific contact resistivity ρ_c $(\Omega \cdot mm^2)$, the concept of transfer length must be introduced.

In practice, current does not flow uniformly across the entire contact pad area, it concentrates at the contact edge, where the metal/semiconductor interface voltage is maximum and decays exponentially with distance from the edge. The transfer length is defined as the distance over which the interface voltage drops to $1/e$ of maximum value, and can be derived as $L_T = \sqrt{\rho_c/R_{sh}}$. In case of $L \geq 1.5 \cdot L_T$, $R_C \approx \frac{\rho_c}{L_T Z} = \frac{R_{sh} L_T}{Z}$, this expression can be substituted to equation (2) become

$$R_{total} \approx \frac{R_{sh}}{Z}(d + 2L_T) \qquad (3)$$

From this equation, the transfer length can be directly extracted from the x-intercept of the $R_{total} - d$ plot, and the specific contact resistivity is then obtained as $\rho_c = R_{sh} \cdot L_T \cdot Z$.

It should be noted that this rectangular TLM extraction method is valid only the mesa width W is equal to contact pad width L ($W \approx L$), otherwise the current would flow across the side edges of the pads. This $W \neq L$ limitation can be avoided with the circular TLM test structures [7].

3.1.2. Measurement Setup

As mentioned earlier, the ohmic metal stack consisting of Ti/Al/Ti/Au (10nm/100nm/45nm/55nm) has been deposited using the electron beam evaporation system followed by Rapid Thermal Annealing (RTA) at 820ºC for 90s in N_2.

TLM measurements were performed on two samples sharing the same ohmic contact metallization scheme (Ti/Al/Ti/Au), differing only in their gate metal deposition – Au gate for Sample 1 and Pt gate for Sample 2. Since the gate metal does not influence the ohmic contact region, the TLM measurements on both samples serve to characterize the ohmic contact and to verify the consistency of the fabrication process between the two samples.

Prior to electrical measurement, the TLM test structures were inspected under an optical microscope to verify the critical geometrical parameters – contact pad width $Z = 300\mu m$ and length $L = 100\ \mu m$ – and to confirm that the mesa isolation was well-defined, ensuring that current does not flow only across the side edges of the pads, satisfying the condition required for accurate TLM extraction. A total of 14 gap spacings were also measured: 2, 4, 6, 8, 12, 14, 16, 18, 21, 24, 26, 28, 31, and 34 µm.

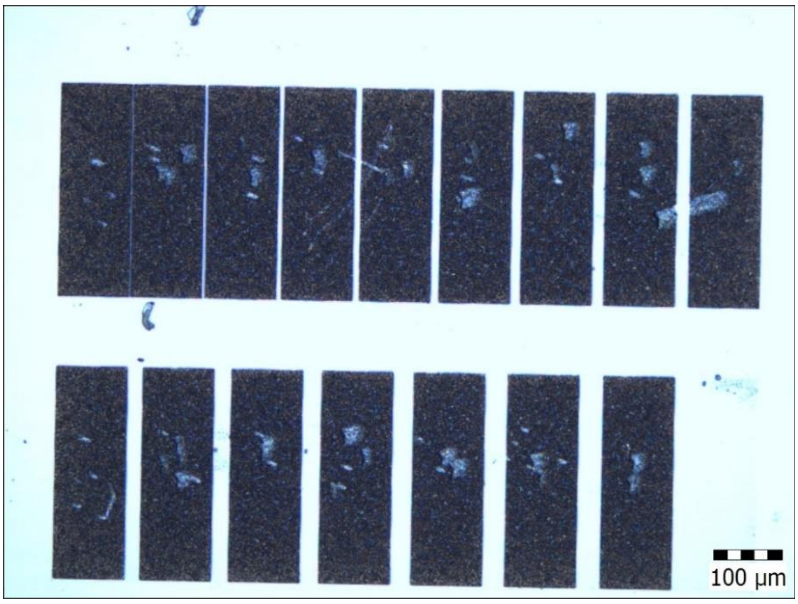

Figure 5. TLM structure on Sample Au Gate deposition

Electrical measurements were performed on both samples using a probe station connected to a Keithley 4250. The pairs of adjacent pads were measured in order of increasing gap spacing with voltage-current (V-I) characteristics sweep in the range of $-500\mu A$ to $+500\mu A$ in source current mode. The ohmic property of the contacts was first confirmed by the linear and symmetric V-I characteristics observed on the Keithley 4250 display. Additionally, the probe contact quality was verified by checking that the measured voltage decreased proportionally with the same applied current across consecutive measurements, ensuring good electrical contact between the probes and the pads.

The total resistance at each gap spacing was extracted from the slope of the V-I curves as $R_{total} = V/I$. Then, the extraction was performed in MATLAB. The R_{total} vs d plot according to the Equation (3) is shown in below figure.

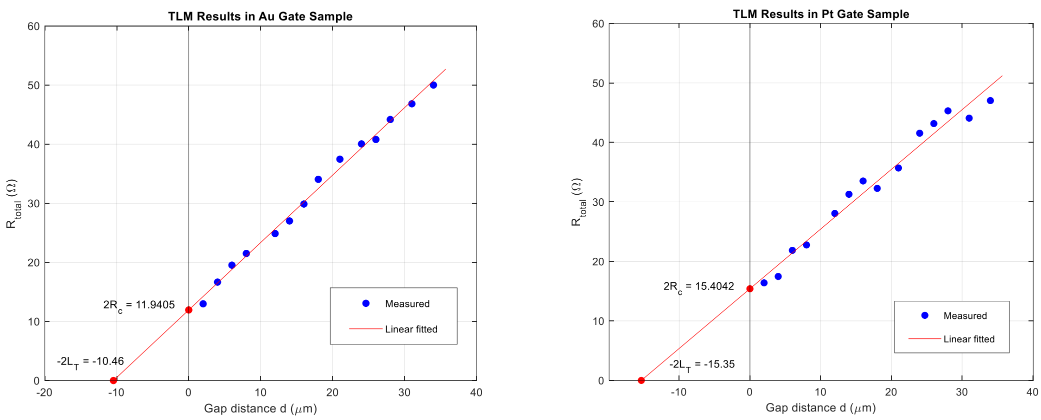

Figure 6. Total resistance vs spacing plot in both samples

3.1.3. Results and Discussions

The extracted parameters of Ohmic contact for both samples are summarized in Table 1. In both cases, the data points of R_{total} vs d plot follow a clearly linear trend, confirming the validity of the TLM model and the ohmic property of the Ti/Al/Ti/Au scheme. The slightly scatters observed in both samples are attributed to the sensitivity to the measurement noise at the low current sweep range of $\pm 500\mu A$ used in this work, and does not significantly affect the validity of the extracted parameters.

	Sample 1 (Au gate)	Sample 2 (Pt gate)
Contact resistance R_C (Ω)	5.97	7.70

Contact resistance $R_C \cdot Z$ $(\Omega \cdot mm)$	1.79	2.31
Specific contact resistance ρ_c $(\Omega \cdot cm^2)$	9.39×10^{-5}	17.7×10^{-5}
Transfer length L_T (μm)	5.23	7.68
Sheet resistance R_{sh} (Ω/sq)	342	300

Table 1. Parameters of Ohmic contact in both samples

The sheet resistance R_{sh} is an important parameter that motivates the use of TLM characterization [7]. The extracted R_{sh} values of $342\ \Omega/sq$ and $300\ \Omega/sq$ for Sample 1 and Sample 2 respectively are of the same order of magnitude, confirming the consistency of the fabrication process and the reliability of the TLM extraction method. These values are also in reasonable agreement with the sheet resistance of $444.2 \pm 3.74\%\ \Omega/sq$ specified in the wafer datasheet. The difference is attributed to the fact that the datasheet value was obtained by a non-destructive eddy current measurement method, which measures the sheet resistance of the grown epitaxial, whereas our measurement was performed after several processes to form ohmic contact, such as annealing – which can slightly modify the 2DEG properties [6].

Although the contact resistance values of both samples are relatively comparable (5.97Ω and 7.70Ω), the specific contact resistivity of Sample 2 ($1.77 \times 10^{-4}\ \Omega \cdot cm^2$) is approximately twice that of Sample 1 ($9.37 \times 10^{-5}\ \Omega \cdot cm^2$). As reported in the literature, the annealing process and surface treatment significantly affect the quality of ohmic contacts [6], [8]. Therefore, the variation observed between the two samples is attributed to minor differences in surface preparation or annealing conditions, rather than a fundamental difference in contact metallization, since both samples use the identical Ti/Al/Ti/Au scheme.

These extracted specific contact resistances agree with values reported in the literature [6] for conventional Ti/Al-based alloyed contacts on AlGaN/GaN, which typically range from 10^{-5} to $10^{-6}\ \Omega \cdot cm^2$ depending on the annealing conditions, the thickness of each metal layer, as well as the AlGaN barrier thickness and Al concentration. Furthermore, the value in order of $10^{-4}\Omega \cdot cm^2$ was also reported in this review article, so our result from Sample 2 is still acceptable.

This metallization scheme was achieved without any advanced process steps, still resulting in acceptable contact resistance for many applications. However, various approaches have been reported to further minimize contact resistance using advanced methods such as recessed ohmic contacts, epitaxial regrowth, ion implantation, and surface treatment, which can improve the specific contact resistance by one to two orders of magnitude [8].

3.2. Schottky Gate Contact

Despite the exceptional properties of AlGaN/GaN HEMTs, Schottky gate contact on AlGaN/GaN heterostructure still suffers from problems such as gate reverse leakage current and fluctuation of the potential barrier at the metal/semiconductor interface. These limitations are commonly attributed to non-ideal metal/semiconductor interface – including surface defects, interfacial layers, and inhomogeneous barrier height [9], [10]. Therefore, detailed characterization of the gate contact is essential to understand the transport mechanisms at the interface limiting device performance.

A Schottky barrier forms inherently at the interface when a metal is brought into contact with an n-type semiconductor, and the ideal SBH is given by Equation (1). In practice, however, the actual SBH rarely reaches this ideal value due to the interface limitations mentioned above. Therefore, the extracted value in many works is referred to as the effective SBH – in contrast to the ideal theoretical value, which only depends on the materials. Despite this deviation, the effective SBH is still highly informative: it reflects the underlying physical phenomena at the metal/semiconductor interface, and also serves as a valuable figure of merit for comparing different gate metallization schemes and guiding further device development.

Various characterization methods have been reported for Schottky gate contact evaluation, including forward I-V characteristics, C-V measurements, and photoemission techniques [7]. Among these, the forward I-V characteristics method is the most fundamental and widely used approach due to its simplicity and direct extraction of key contact parameters [7], [11]. In this work, the Schottky gate contact is characterized using the thermionic emission model applied to forward I-V measurements. The current through the Schottky gate is described by conventional diode equation and thermionic emission equation [12]:

$$I = I_S \cdot \left[\exp\left(\frac{q(V - IR_S)}{nkT}\right) - 1\right] \qquad (4)$$

$$I_S = A_{eff} \cdot A^* \cdot T^2 \cdot \exp\left(\frac{-q\Phi_b}{kT}\right) \qquad (5)$$

where V is the applied voltage, I_S is the reverse saturation current, A_{eff} is effective area of the junction, A^* is the Richardson constant, T is temperature, k is Boltzmann constant, R_s is series resistance of semiconductor, Φ_B is SBH, and n is ideality factor. The ideality factor n describes how closely the junction as an ideal Schottky diode – for an ideal, $n = 1$, indicating that thermionic emission is the sole transport mechanism. Values of $n > 1$ indicate the presence of additional transport mechanisms at the interface, such as interface traps, tunnelling, barrier inhomogeneity, or recombination currents.

Since the gate functions as a rectifying contact that must suppress reverse leakage current, a high SBH is expected. According to Equation (1), this is achieved by selecting metals with a high work function. In this work, Au and Pt with work functions of 5.1eV and 5.65eV [13], respectively, are investigated as gate metals. In addition to their high work functions, both Au and Pt possess low resistivity, which minimizes the gate resistance, contributing the higher cut-off frequency and maximum frequency [14].

3.2.1. Circular TLM test structure

Characterizing the Schottky gate contact directly on a transistor is impractical, as the measured I-V characteristics would reflect the combined effects of multiple parameters – such as gate-source spacing L_{gs}, gate-drain spacing L_{gd}, etc. – making it difficult to isolate the gate contact parameters alone. A dedicated test structure is therefore used to characterize the Schottky gate contact and evaluate it independently.

As discussed in Chapter 3.1.1, the circular TLM (CTLM) structure eliminates the mesa isolation requirement ($W = Z$) of the standard TLM. In this work, the same principle is applied for gate characterization – the CTLM pattern is no longer used to extract contact resistance, but rather as a set of circular lateral Schottky diodes with different Schottky gate diameters, as shown in Figure 7. The inner circle serves as the Schottky gate contact, while the outer ring serves as the ohmic contact, ensuring that all measured current flows through the Schottky junction.

The use of multiple diodes with different gate diameters provides an additional advantage – it allows verification of the structural uniformity across the sample by comparing the density current – voltage (J-V) characteristics, where $J = \frac{I}{A_{eff}}$, the J-V curves of all diameters should overlap if the Schottky contact is ideally uniform.

3.2.2. Parameter Extraction Methods

In conventional vertical Schottky diode structures – such as metal contacts on highly-doped n-type GaAs or GaN – the series resistance R_s is typically negligible, and the effective area A_{eff} is well defined. In such cases, the thermionic emission equations are simplified directly to [7]

$$J = J_S \cdot \left[\exp\left(\frac{qV}{nkT}\right) - 1\right] \qquad (6)$$

$$J_S = A^* \cdot T^2 \cdot \exp\left(\frac{-q\Phi_b}{kT}\right) \qquad (7)$$

when applied voltage $V \gg \frac{kT}{q}$, Equation (6) can be described as

$$\ln(J) = \ln(J_S) + \frac{q}{nkT}V \qquad (8)$$

From Equation (8), the ideality factor n and the saturation current density J_S can be directly extracted from the slope and y-intercept of the ln(J)-V plot, respectively. From extracted J_S, the SBH Φ_B is determined by Equation (7). The Richardson constant $A^* = 26.4\frac{A}{cm^2K^2}$ for AlGaN/GaN heterostructure [15]. This method can be called purely thermionic emission (TE) extraction method.

However, in the context of the lateral structure in this work, with unintentionally doped epitaxial layers, the series resistance R_S is considerable. Because when the current magnitude increases, the voltage drop IR_s across the series resistance becomes non-negligible in Equation (4). Therefore, in this regime, the purely TE method is no longer reliable, and a modified extraction approach is required. In this work, Cheung's method [12] is employed to account for the series resistance contribution. In case of $V \gg \frac{kT}{q}$, by rearranging Equation (4) and (5), we have

$$V = R_s I + n\Phi_B + \frac{nkT}{q}\ln\left(\frac{I}{A_{eff}A^*T^2}\right) \qquad (9)$$

Derivative Equation (9) with respect to ln (I):

$$\frac{dV}{d(lnI)} = R_s I + \frac{kT}{q}n \qquad (10)$$

By plotting $dV/d(lnI)$ vs I curve, the ideality factor n and the series resistance R_s can be extracted from the y-intercept and the slope, respectively.

From Equation (9), defining the function $H(I) = V - \frac{nkT}{q} \ln \left(\frac{I}{A_{eff}A^{*}T^{2}}\right)$, we have:

$$H(I) = R_s I + n\Phi_B \qquad (11)$$

By plotting $H(I)$ vs I curve and finding y-intercept, with extracted ideality factor n from Equation (10), the SBH Φ_B is determined. Additionally, the series resistance R_s can again be extracted from Equation (11) and compared to the value obtained from Equation (10) to validate this extraction method.

It should be noted that Cheung's method has two limitations. First, the numerical differentiation in Equation (10) amplifies measurement noise, particularly at low current levels where the SNR is small – making the $dV/d(lnI)$ vs I plot less reliable. Second, in the lateral structure used in this work, the effective junction area A_{eff} is approximated as the inner circle area $\frac{\pi d^2}{4}$, however, since the current flows not only vertically through the junction but also laterally, this approximation introduces non-uniformity in the absolute current density distribution [15]. Nevertheless, within the scope of this work, this deviation is considered negligible and the extracted parameters are treated as representative values for comparative purposes.

3.2.3. Measurement Setup

In this work, Au (150nm) and Pt (100nm) gate contacts were deposited by e-beam evaporation. Prior to gate deposition, a reactive ion etching (RIE) surface treatment was applied to the Sample 1 of Au Gate to clean the AlGaN surface, while no such surface treatment was applied to Sample 2 of Pt Gate.

Forward I-V measurements were performed on the CTLM Schottky gate structures of both Sample 1 (Au gate) and Sample 2 (Pt gate) using the same probe station and Keithley 4250 as used for the TLM measurements. Two sets of measurements were performed on each diode to cover the full forward bias range and get high data-point density. Set 1 covered the low-to-moderate bias regime (0–2.5V) using source voltage mode, providing sufficient resolution for TE parameter extraction in this region. Set 2 covered the high bias voltage regime (2.5–5V) using source current mode, providing sufficient resolution for Cheung's method extraction in this region.

Not all diodes were included in this work. Diodes exhibiting visible short-circuit on the surface were excluded from the measurements. Furthermore, the diodes with largest inner circle diameters were also excluded, as the small gap ($\sim\mu m$) between the inner gate contact and the outer ohmic may increase the risk of leakage current – either through gate metal diffusion or lateral surface conduction – resulting in the measurements less reliable.

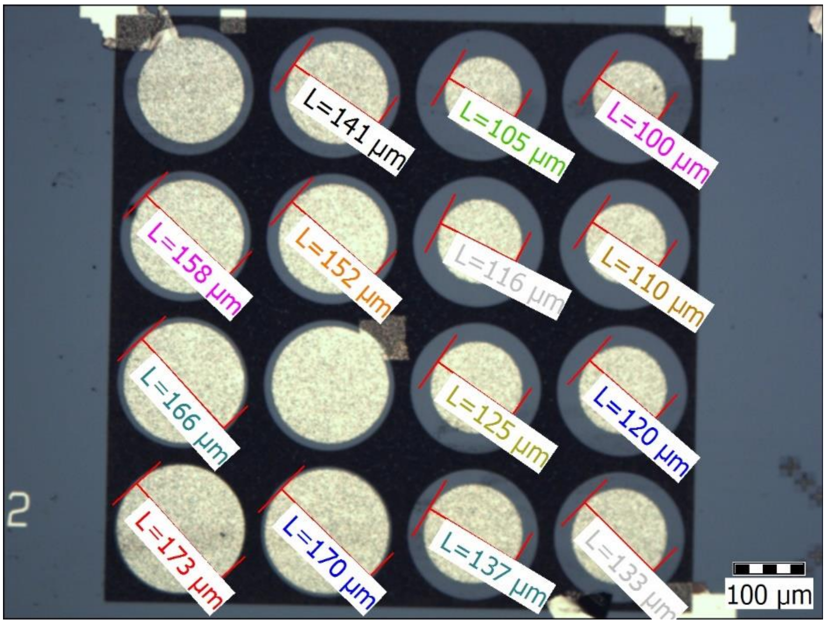

Figure 7. CTLM pattern in Sample Au Gate

3.2.4. Results and Discussions

First, we will investigate the Sample 1 of Au gate. As shown in Figure 8, the J-V characteristics of measured diodes exhibit good uniformity, confirming consistent Schottky contact formation. All curves share a common trend characterized by three distinct regions. In the first region, the current follows an ohmic relationship governed by the shunt resistance R_{sh}, as expressed by $I = \frac{V}{R_{sh}}$ [16]. In the second region, the current increases exponentially, a behaviour characteristic of a Schottky junction, indicating that thermionic emission is the dominant transport mechanism [16]. The series resistance effect is negligible, and the purely TE extraction method is therefore applied here. In the third region, the rate of current increase slows down, indicating the growing contribution of series resistance R_s at high forward bias as we discussed before. Cheung's method is therefore applied in this region to extract R_s alongside n and Φ_B.

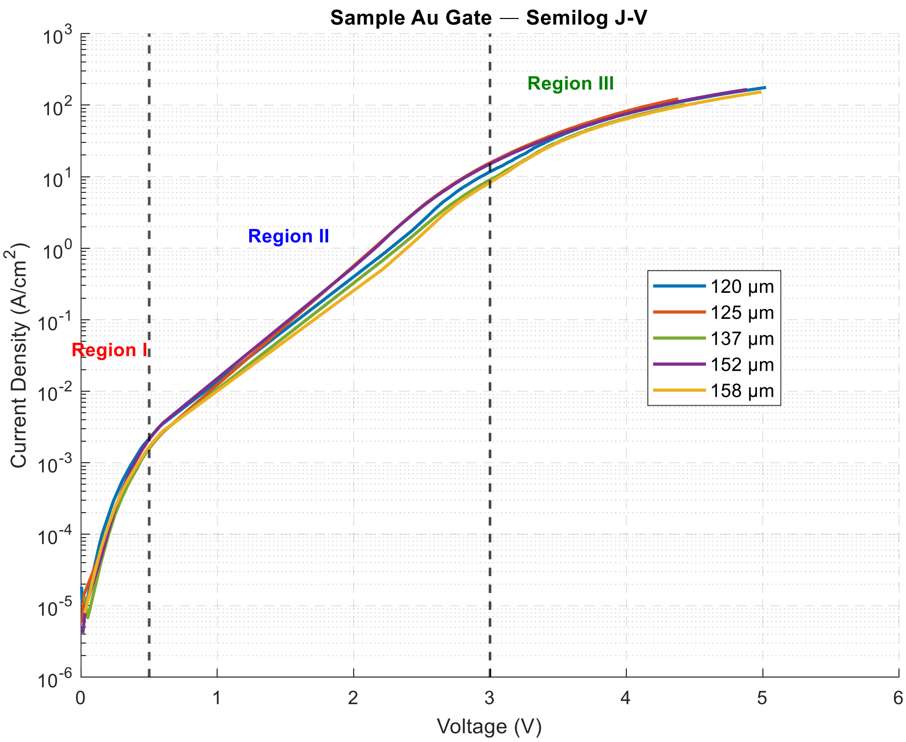

Figure 8. Density current - Voltage plot of Sample Au Gate

The purely TE extraction method is applied in Region II. The mathematical extraction was performed in MATLAB. The extracted effective SBH Φ_B and ideality factor n are summarized in Table 2. The validity of this extraction is confirmed by the agreement between the measured J-V curve and the calculated curve reconstructed from the extracted parameters, as shown in Figure 9.

Diameter (μm)	Ideality factor n	SBH Φ_B (eV)
120	11.88	0.58
125	11.69	0.58
137	11.56	0.58
152	12.58	0.57
158	12.39	0.58

Table 2. SBH and ideality factor of Au gate contacts obtained from purely TE method in region 2

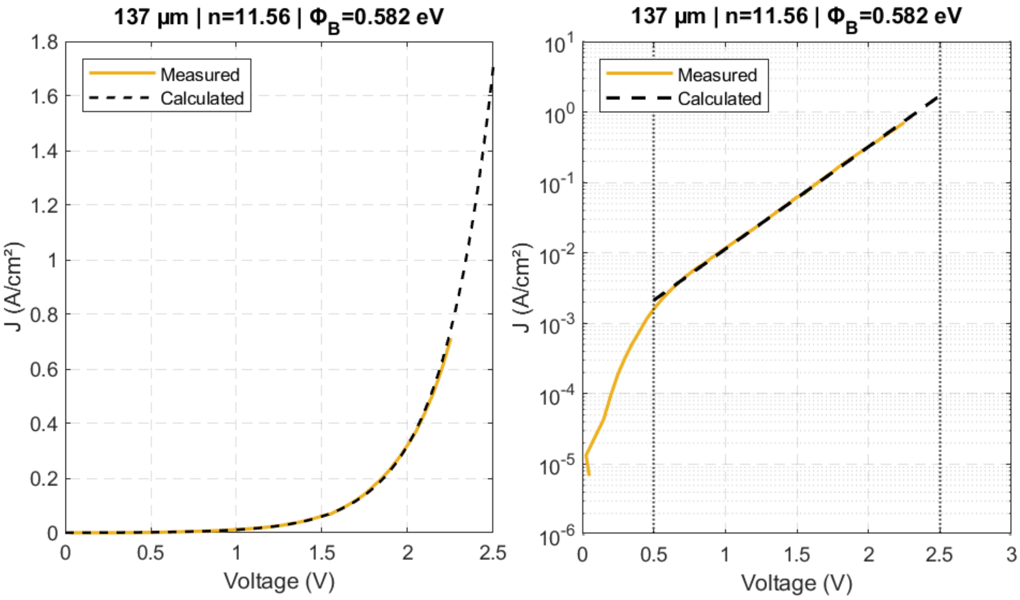

Figure 9. The measured and calculated J-V curves of 137μm Au gate diode in linear and log scale

Despite the good uniformity observed among the different Au gate contacts and the agreement between the calculated and measured J-V curves, the extracted SBH and ideality factor deviate significantly from the expected values: $\Phi_B = 0.9 - 1.3eV$ and $n = 1 - 2$ for Au contacts [17], [18]. While ideality factors slightly above 2 can be attributed to barrier height inhomogeneity at the metal/semiconductor interface [19], the extracted values of $n = 12$ in this work are considerably higher, suggesting additional unknown non-ideal mechanisms. These results are likely attributed to several factors: non-ideal surface conditions prior to gate metal deposition, impurities in the deposited metal, or the limitation of applying the purely TE extraction method in the chose voltage range.

In region III, the Cheung's method is applied because of series resistance, and the extracted parameters are compared to those obtained from the purely TE method in Region II. The mathematical extraction was performed in MATLAB. The extracted SBH Φ_B, ideality factor n, and series resistance R_s are summarized in Table 3. The $\frac{dV}{dlnI}$ $vs\ I$ plot and $H(I)$ $vs\ I$ plot are also shown in Figure 10. The validity of this extraction is confirmed by the series resistance extracted from two plots, as well as the agreement between the measured and the calculated J-V curve reconstructed from the extracted parameters, as shown in Figure 10.

Diameter(μm)	Ideality factor n	SBH Φ_B (eV)	Series resistance $R_s(\Omega)$ from $H(I)$ vs I plot	Series resistance $R_s(\Omega)$ from $dV/d(lnI)$ vs I plot

120	11.11	0.58	67.21	67.00
125	14.18	0.51	47.73	47.85
137	9.20	0.64	60.96	61.43
152	14.10	0.52	36.97	37.03
158	11.06	0.59	42.28	41.90

Table 3. Parameters of Au gate contacts obtained from Cheung's method in region 3

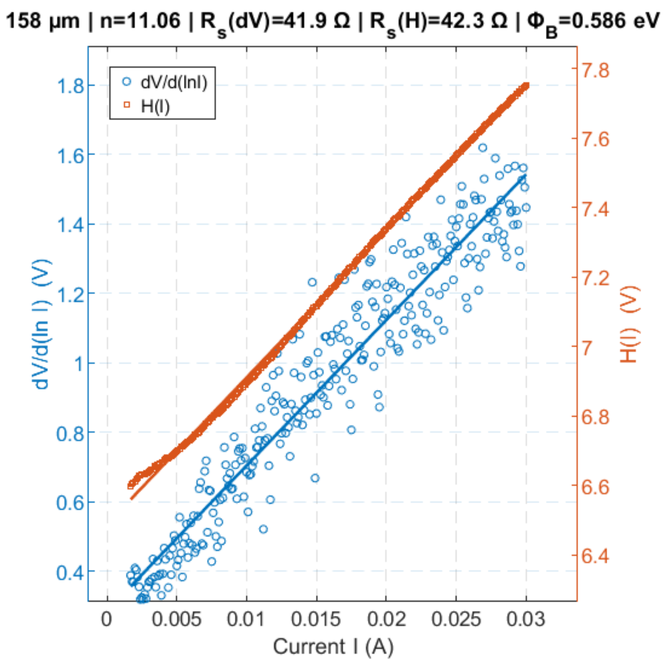

Figure 10. The dV/dlnI vs I plot and H(I) vs I plot

The results obtained from Cheung's method are subject to the sensitivity of the numerical differentiation in the $\frac{dV}{dlnI}$ $vs\ I$ plot to measurement noise. However, with current levels in the range of 5–30mA measured in this range, the SNR is considered sufficiently high [10], suggesting that noise amplification alone does not fully explain the observed nonlinearity of this plot. Nevertheless, the extracted results from Cheung's method remain unreasonable, consistent with the results obtained from the purely TE method, confirming that the non-ideal behaviour is not only due to the extraction method.

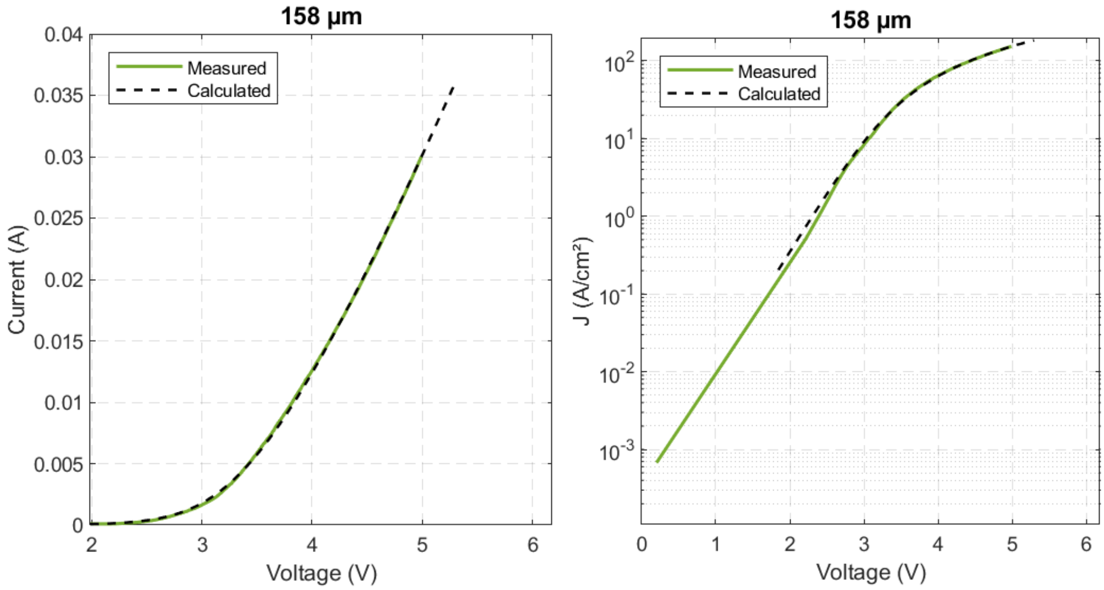

Figure 11. The measured and calculated J-V curves of 158μm Au gate diode in linear and log scale

The too high ideality factors and too low SBHs from both extraction methods may be attributed to one or more of the following factors: non-ideal metal/semiconductor interface quality; impurity metal; unknown or mixed transport mechanisms, such as trap-assisted tunnelling, barrier inhomogeneity, or shunt conduction, that deviate from the pure thermionic emission model assumed in both methods; an inappropriate choice of voltage range for parameter extraction, where the boundaries between the three transport regions are not sharply defined. A more complete understanding would require more dedicated characterization, such as C-V measurements, temperature-dependent I-V measurements, or more sophisticated extraction approaches, such as the two-diode model [11] or numerical solution methods [20], which explicitly account for mixed transport mechanisms and barrier inhomogeneity.

In the case of Sample Pt Gate, the J-V characteristics exhibit significant non-uniformity among the measured gate contacts – with considerable variation in the threshold voltage, slope, and current density magnitude across different diameters. This level of non-uniformity does not provide a reliable basis for meaningful parameter extraction, and the Schottky characterization of Sample 2 is therefore not pursued further. This behaviour is attributed to the absence of RIE surface treatment prior to Pt gate deposition in Sample 2, in contrast to Sample 1 where RIE was applied. The native oxides and surface contaminants remaining on the AlGaN surface likely led to a non-ideal and spatially non-uniform metal/semiconductor interface, resulting in inconsistent Schottky contact formation across the sample. This result highlights the critical role of surface preparation in determining the quality of the Schottky gate contact – even when all other fabrication parameters, including the gate metal work function, deposition method, and epitaxial structure, are comparable between the two samples.

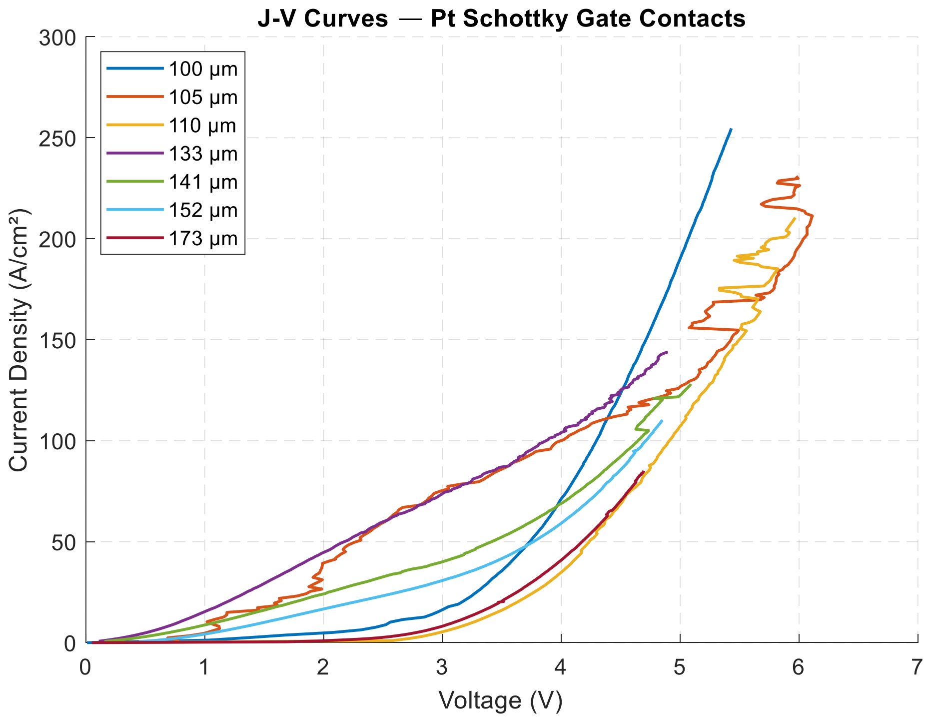

Figure 12. Density current - voltage in Sample Pt gate

4. Conclusion

In this work, the ohmic and Schottky gate contacts of AlGaN/GaN HEMTs were electrically characterized using TLM and CTLM test structures. Two samples with identical structure but different gate metals and surface treatment – Au (Sample 1) with RIE and Pt (Sample 2) without RIE – were investigated to assess the influence of both gate metal choice and surface treatment (RIE) choice on contact properties.

The TLM extraction gave sheet resistance and specific contact resistivity values consistent with the expected range for AlGaN/GaN heterostructures, confirming the reliability of the ohmic contacts and the consistency of the fabrication process between the two samples.

For the Schottky gate characterization, three distinct transport regions were identified from the forward J-V characteristics. In Sample 1 (Au), the purely TE method and Cheung's method were applied in the thermionic emission and series resistance dominated regions, respectively. Both methods consistently gave non-ideal results – too high ideality factors (~12) and low SBHs (~0.6) – attributed to non-ideal interface quality, deposited metal or the limitation of applied method in lateral CTLM pattern. The significant non-uniformity observed in Sample 2 (Pt) is attributed to the absence of RIE surface treatment, highlighting the critical role of surface preparation in Schottky gate contact quality.

For future work, optimizing the surface treatment, metal deposition processes, performing T-I-V and C-V measurements, and applying more sophisticated extraction methods such as the two-diode model are recommended to gain a deeper understanding of the non-ideal gate contact behaviour observed in this work. Once a well-characterized and reliable Schottky gate contact is established, the influence of additional gate related parameters, such as gate length and source/drain-to-gate spacing, on transistor performance can be investigated through standard three-terminal measurements, including output characteristics (I_D vs V_{DS}) and transfer characteristics (I_D vs V_{GS}).

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